SiPM statistical response: validation pilot

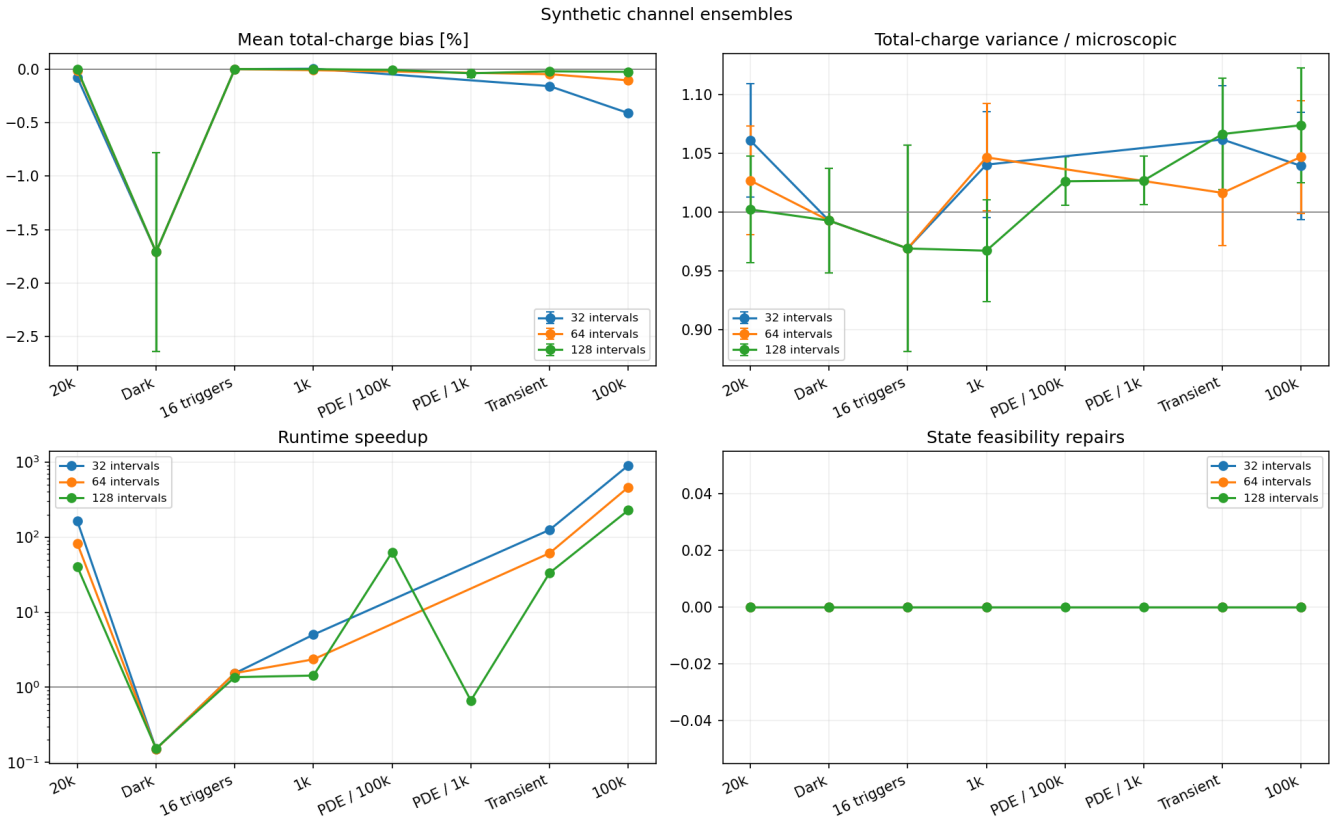
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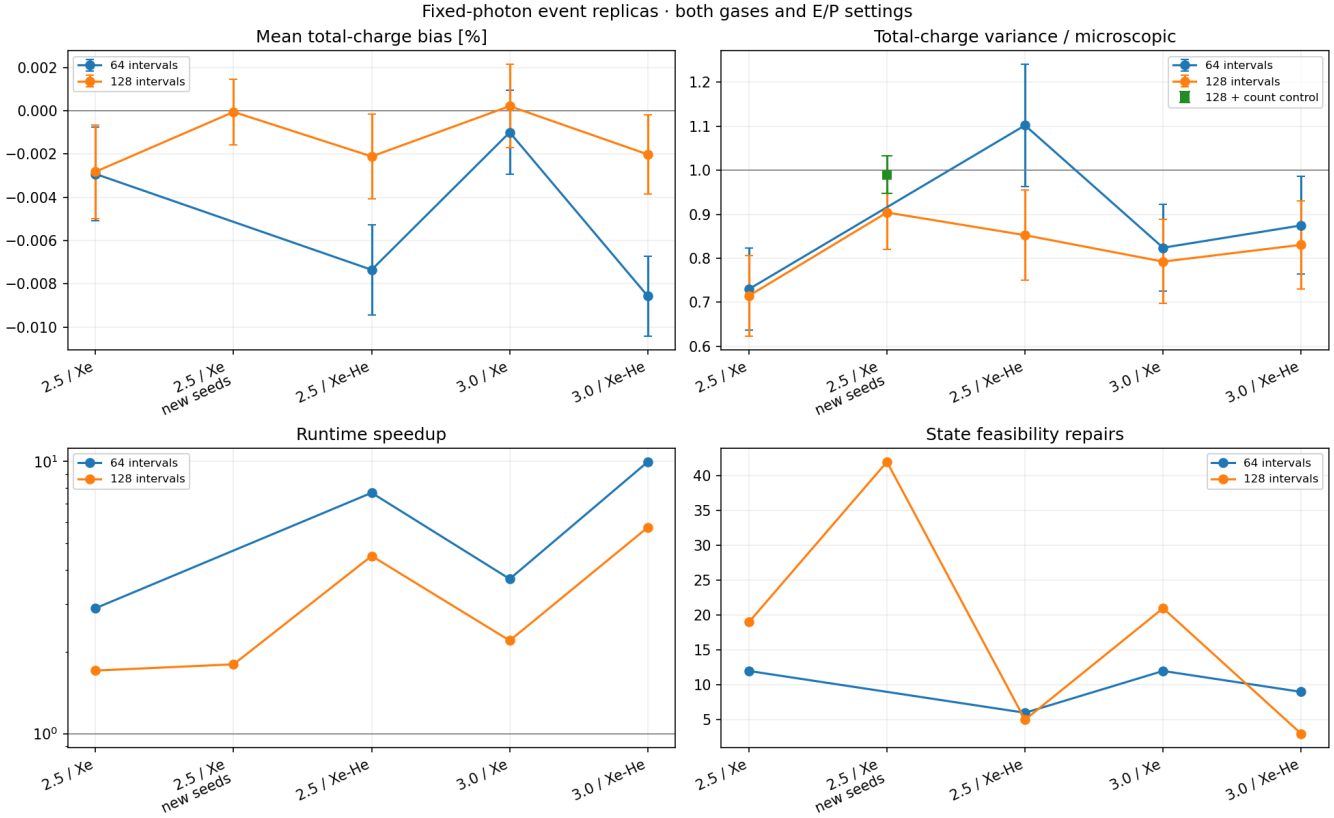
Default: 128 intervals per 250 ns bucket. Maximum saved-event mean-charge difference: 0.0028%. Measured sensor speedup: 1.7 to 5.7 times. The two-event pilot completed for both gases, including the ASIC stage. Full production remains stopped.

The statistical model preserves binomial PDE and Poisson dark-count sampling. Fixed counts are partitioned into short intervals; finite-cell occupancy gives conditional charge moments. A bounded beta draw supplies charge fluctuations, and the same charge updates recovery. The second recovery moment uses a closure preserving its conditional expected loss. Sparse buckets use individual arrival times and aggregate-state draws. The physical setup has 7,854 cells, 73.4 ns recovery and 250 ns buckets. Synthetic optical cases use PDE = 1 and zero dark rate to isolate response fluctuations; the two additional PDE cases use PDE = 0.30 and 10,000 replicas. Real-event replicas use PDE = 0.30 and the production dark rate, with the same fixed photon input in each replica. The first bb0nu event of each of the four photon files (both gases at E/P = 2.5 and 3) is repeated 256 times. The E/P = 2.5 pure-Xe case is also checked with 512 independent new seeds. Other synthetic cases use 2,000 replicas, except sparse pulses (20,000). The card selects 128 intervals; 32 and 64 measure discretization sensitivity. Error bars show one Monte Carlo standard error. Pilot review budgets are 0.05% mean discrepancy and 5% variance discrepancy, or four Monte Carlo standard errors when larger. Covariance uses four standard errors plus 5% of the neighbouring standard-deviation product. Synthetic table checks evaluate each bucket and its adjacent covariance. Real-event checks evaluate total-event charge; covariance is marked N/A. Plots show total-channel or total-event charge, conditional on fixed photon input. This spread is the sensor contribution, not the full event energy resolution. Sensor-computation speedups use median time per real-event replica; synthetic speedups use the complete ensemble runtime. Feasibility repairs and bounded-variance limiting are counted explicitly.

Independent-seed event check: charge variance ratio = 0.904 +/- 0.084 (one standard error). Detected-count variance / analytical binomial variance: microscopic 1.085 +/- 0.073; statistical 0.952 +/- 0.059. Using the known binomial second moment as a control variate gives charge-variance ratio 0.990 +/- 0.043. The variance estimator subtracts 0.5 times the difference between the sampled squared detected-count deviation from its known mean and its analytical variance. That subtraction has expectation zero; the fixed coefficient is the same for both models. Standard errors come from the per-replica estimator contributions.

At 128 intervals, 90 second-state-moment repairs occurred in 1,445,328,831 aggregate draws; the largest correction to S was 0.1484. Variance-limit adjustments: 0. State repairs adjust the dimensionless second recovery sum S.


Synthetic channel ensembles
Mean total-charge bias [%]
Total-charge variance / microscopic
Runtime speedup
State feasibility repairs
32 intervals
64 intervals
128 intervals
20k
Dark
16 triggers
1k
PDE / 100k
PDE / 1k
Transient
100k


Fixed-photon event replicas · both gases and E/P settings
Mean total-charge bias [%]
Total-charge variance / microscopic
Runtime speedup
State feasibility repairs
64 intervals
128 intervals
128 + count control
2.5 / Xe
2.5 / Xe new seeds
2.5 / Xe-He
3.0 / Xe
3.0 / Xe-He

Case	Bias (%)	Var. ratio	Speedup	Mean	Variance	Covariance	Repairs	Limits
events_ep2p5_pure_xe_independent_steps_128	-6.023e-05	0.9041	1.798	pass	pass	N/A	42	0
events_ep2p5_pure_xe_steps_128	-0.002823	0.715	1.706	pass	pass	N/A	19	0
events_ep2p5_pure_xe_steps_64	-0.002926	0.7301	2.89	pass	pass	N/A	12	0
events_ep2p5_xe_he15_steps_128	-0.002117	0.8526	4.497	pass	pass	N/A	5	0
events_ep2p5_xe_he15_steps_64	-0.007354	1.102	7.671	pass	pass	N/A	6	0
events_ep3_pure_xe_steps_128	0.0002082	0.7925	2.198	pass	pass	N/A	21	0
events_ep3_pure_xe_steps_64	-0.001003	0.8241	3.702	pass	pass	N/A	12	0
events_ep3_xe_he15_steps_128	-0.002023	0.8307	5.726	pass	pass	N/A	3	0
events_ep3_xe_he15_steps_64	-0.00857	0.8747	9.967	pass	pass	N/A	9	0
synthetic_bright_steps_128	-0.001896	1.002	40.33	pass	pass	pass	0	0
synthetic_bright_steps_32	-0.07692	1.061	162.6	FAIL	pass	pass	0	0
synthetic_bright_steps_64	-0.0114	1.027	81.97	pass	pass	pass	0	0
synthetic_dark_steps_128	-1.711	0.9927	0.153	pass	pass	pass	0	0
synthetic_dark_steps_32	-1.711	0.9927	0.151	pass	pass	pass	0	0
synthetic_dark_steps_64	-1.711	0.9927	0.1501	pass	pass	pass	0	0
synthetic_few_steps_128	0.002137	0.969	1.366	pass	pass	pass	0	0
synthetic_few_steps_32	0.002137	0.969	1.536	pass	pass	pass	0	0
synthetic_few_steps_64	0.002137	0.969	1.545	pass	pass	pass	0	0
synthetic_moderate_steps_128	0.001267	0.9671	1.435	pass	pass	pass	0	0
synthetic_moderate_steps_32	0.005812	1.04	5.026	pass	pass	pass	0	0
synthetic_moderate_steps_64	-0.008312	1.047	2.359	pass	pass	pass	0	0
synthetic_pde_bright_steps_128	-0.005608	1.026	63.11	pass	pass	pass	0	0
synthetic_pde_moderate_steps_128	-0.03709	1.027	0.6663	pass	pass	pass	0	0
synthetic_transient_steps_128	-0.01861	1.066	33.35	pass	pass	pass	0	0
synthetic_transient_steps_32	-0.1573	1.062	124.8	FAIL	pass	pass	0	0
synthetic_transient_steps_64	-0.0453	1.016	61.15	FAIL	pass	pass	0	0
synthetic_very_bright_steps_128	-0.02377	1.074	228	pass	pass	pass	0	0
synthetic_very_bright_steps_32	-0.4096	1.039	899.1	FAIL	pass	pass	0	0
synthetic_very_bright_steps_64	-0.1033	1.047	461	FAIL	pass	pass	0	0